popquiz kine1D-3

This is a preview of the published version of the quiz

Started: Apr 29 at 7:27pm

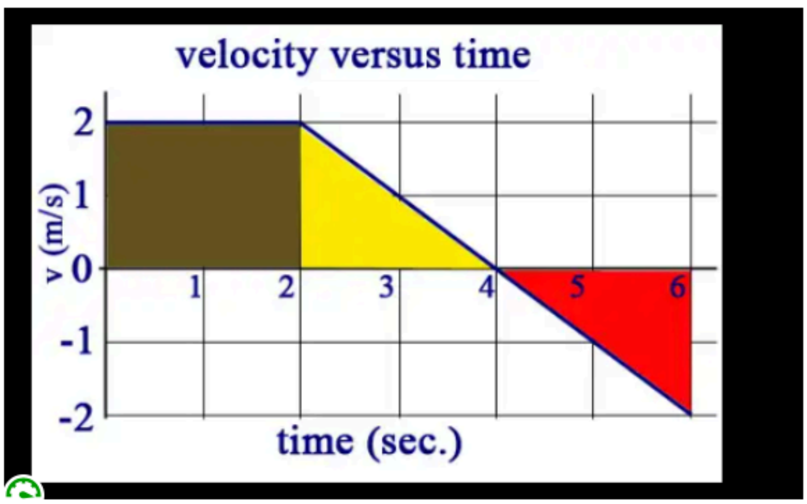
Quiz Instructions

one attempt. No make up

magnitude = does not include direction. A magnitude can not be negative.

Question 1 20 pts

Here is a graph showing the veocity of an object as a function of time. On the number line.



What is the total displacement deltaX of the object:

hint: displacement is the integral of the V(t) = area under the graph. **but !!!!! remember calculus. the area below the x-axis is negative (in red).** There is a subtraction here. You should all the know the formula for area of triangle and rectangle.

- ○ 2m
- ○ 4m
- ○ 0m

○
8m

Question 2 20 pts

Building on previous question (same graph)

When is the object stopping? (at what time it is at rest

○
t=6s

○
t=4s

○
Its does not stop. Its on the move the all time

○
t=2s

Question 3 20 pts

Building on the previous question,

Between 2s and 4 s, the car is moving

(hint: for the direction-> its the sign of the velocity that is important. Check y-axis

for the magitude (how fast) its the numbers that matters. are numbers on the y-axis increase or decrease in magnitude.).

○
@ right and at constant velocity

○
@ right and slowing down

○
@ left and speeding up

○
@ left and slowing down

○
@ right and speeding up

Question 4 20 pts

building on the same graph.

Between 4s and 6 s , the car is moving

hint: again. Observe the y-axis. is the velocity negative or positive? is the magnitude of the velocity (ignore here the negative sign). Is it increasing of decreasing?

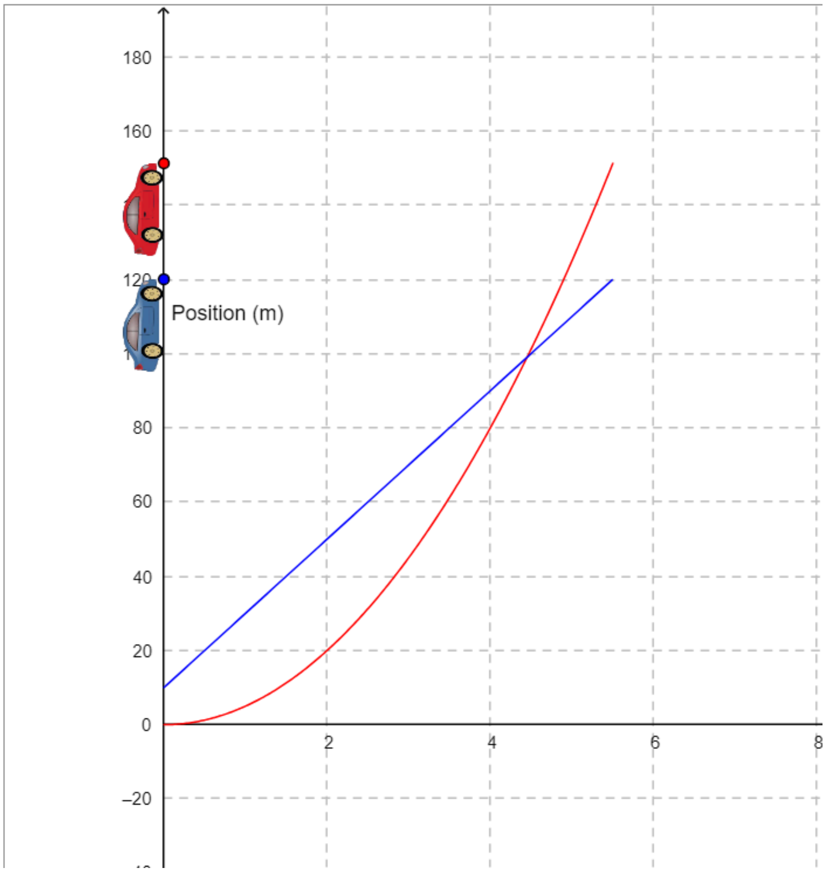
- @ left and slowing down
- @ right and speeding up
- @ constant speed
- @ left and speeding up

Question 5 20 pts

What is the acceleration of the car between 2 and 4 s?

- 0m/s/s
- -1m/s/s
- 1m/s/s
- -2m/s/s

Question 6 5 pts



bonus. at t=0, x=0 the chase starts. You see that the red car starts from rest (Vo=0 at t=0) and accelerates at a constant rate. Lets try to find its acceleration. at t=2s evaluate its instantaneous velocity (slope =tangent)

pay attention to the scales on the axis

- ○ 60m/s
- ○ 40m/s
- ○ 5m/s
- ○ 20m/s

Question 7 5 pts

Building on the previous question. The acceleration of the red car is ?

(between time t=0 and t=2s you have the change in speed)

- ○ 30m/s/s
- ○ 10m/s/s

- 2.5m/s/s
- 20m/s/s

Question 8 5 pts

What is the equation of motion of the red car

- x= 10t + 10t^2
- x= 20t +5t^2
- x= 20 + 10 t^2
- x=5t^2

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